

A Combination of Visual-Semantic Reasoning and Text Entailment-based Boosting Algorithm for Cheapfake Detection

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ABSTRACT

Misuse of real photographs with conflicting image captions in news items is one case of out-of-context (OOC) misuse of media. To detect out-of-context given pair of news (i.e., captions) and attached image, people should determine the truthfulness of the statement and evaluate whether the triplet talks to the same event. This paper presents a new method to detect the OOC media challenge introduced in ACMMM'22 Grand Challenge on Detecting Cheapfakes. For *Task₁* (i.e., detect conflicting image-caption triplets), our approach uses bottom-up attention with visual semantic reasoning to extract global features of the image to perform comprehensive image-text matching, multiple models of natural language processing to extract semantic relation of the caption and utilize boosting to improve the accuracy of the performance. Our method achieved an 85.5% accuracy score on 80% of the testing dataset and has 4.5% higher accuracy than baseline. For *Task₂* (i.e., determine whether a given (Image, Caption) pair is real or fake), we detect the veracity of captioned image base semantic features and correlation between image/caption and achieve 76% accuracy. Our source code is available at https://github.com/latuanvinh1998/Cheapfakes_detection_acmmm. Docker for submission is available at <https://hub.docker.com/repository/docker/latuanvinh1998/acmmmcheapfakes>.

CCS CONCEPTS

• Computing methodologies → Natural language processing; Computer vision.

KEYWORDS

Deep Learning, Computer Vision, Natural Language Processing, Image-Text Matching, Cheapfakes, Misinformation, News

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1 INTRODUCTION

The rapid development of social media has recently increased public information and news and made it easy to acquire and get knowledge. It also increases the quantity and type of misinformation. Prior research focuses primarily on fake news detection, fact verification, and deepfake detection. Fazio et al. [7] have demonstrated that out-of-context (OOC) captioned images (i.e., alteration of context) are an effective way of misinformation and are more prevalent and easy to make. Besides, images also increase the interaction and spreading of news, article, or post on social media. Technologyreview¹ also reports that cheap fakes cause more damage than deepfake at the election in 2020.

Detecting fake news is challenging, but detecting cheapfakes is even more complicated. People must either detect the statement's veracity or verify whether the news and attached image correlate. Besides, determining the relation between text and photos in the information is difficult. Unlike common tasks in joint computer vision and natural language processing like image-text matching or image captioning, the association between given text and caption in the news domain is tough to judge. Sometimes, there is abstract relation between them, and it is tough to even for humans. To address these issues, ACMMM'22 Grand Challenge on Detecting Cheapfakes [3] challenges participants to detect whether given samples (news image and associated captions) are out-of-context.

Recently, Aneja et al. [2] presented a method that uses a matching caption to the corresponding image region and combines the semantic similarity of two captions to detect whether the pair (image, caption₁, caption₂) is out-of-context. This method's weakness is its inability to compare the relation level between image and

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¹<https://www.technologyreview.com/2020/12/22/1015442/cheapfakes-more-political-damage-2020-election-than-deepfakes/>

caption. In reality, the caption usually mentions multiple objects or the entire image. Hence, cosine similarity may not be strong enough to express the semantic correlation between two captions.

Based on the abovementioned discussion, we propose a new method to address the issues mentioned by combining Visual-Semantic Reasoning and Text Entailment-based Boosting Algorithm. The rest of this paper is to describe our method and evaluate it using datasets and baseline introduced in the overview paper of the grand challenge.

2 METHODOLOGY

2.1 Task 1

First, we utilize Visual Semantic Reasoning [13] for training image-caption matching tasks on training and validation data. We also apply the pre-trained of WK-SBERT [15] on sentence similarity task and DeBERTa [9] on MultiNLI task [16] to extract semantic correlation between caption in comprehensive way. Next, to combine outputs from multiple pre-trained models, we use boosting [8] on the first 200 captioned images in the testing dataset to improve accuracy. We evaluate the performance of our method on the rest 800 captioned image.

2.1.1 Image-Text Matching. As mentioned, the baseline [2] lack of ability to measure the correlation between caption and the global context of the image. Throughout our experience, we recognize that using features extracted from Mask-RCNN is not enough to express the image's content; and a caption mostly mentions multiple objects in an image. To overcome these issues, we propose Visual Semantic Reasoning [13] for the image-text matching approach, in which output is value to determine the semantic relation of a given captioned image.

To extract region features of images, we use bottom-up-attention with Faster-RCNN [14] which is pre-trained on Visual Genome [11]. Since the Visual Genome dataset has diverse classes such as 'sky,' 'water,' and 'grass,' image's content can be presented with more semantic information. Inherited from VSRN paper [13], we set the confidence threshold for each image at 0.2 and the IOU threshold at 0.7 and chose the top 36 ROIs with the highest class detection score. Each image can be represented as set of features $V = \{v_1, \dots, v_k\}$, $v_i \in \mathcal{R}^D$, where D is 2048.

To extract global features and perform image-caption matching on training and validation data, we apply Visual Semantic Reasoning for Image-Text Matching techniques [13].

To extract the representation of the caption with the same dimensionality as the global features of the image, we run GRU Encoder [4]. Before running the GRU Encoder, we apply Named Entity Recognition (NER) to generalize entities in the caption to their corresponding type (Figure 1), which reduces the abstraction relationship between image and caption.

For objective function, we take advantage of hinge-based triplet ranking loss [12],[13],[6] and choose negative sample which have closet margin to each training sample. See Eq.1.

$$\mathcal{L}_M = [\alpha - S(I, C) + S(I, \hat{C})]_+ + [\alpha - S(I, C) + S(\hat{I}, C)]_+, \quad (1)$$

We define α is margin parameter, $[loss]_+ \equiv \max(loss, 0)$, S as dot similarity function, $\hat{C} = \operatorname{argmax}_{C' \neq C} S(I, C')$ and $\hat{I} = \operatorname{argmax}_{I' \neq I} S(I', C)$

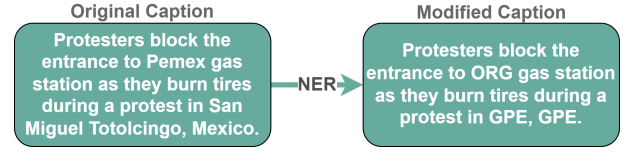


Figure 1: Text pre-processing with NER(Spacy).

as hardest negative sample for a matching pair (I, C) . We choose the hardest negative sample in each mini-batch to reduce computation instead of the entire training set.

In our experience, making us of log-likelihood of the predicted output sentence will return a higher retrieval score on the validation set instead of using triplet ranking loss solely. The objective function aims to generate sentences similar to the ground truth caption. The loss function is defined as follows

$$\mathcal{L}_G = - \sum_{i=1}^N \log(y_t | y_{t-1}, V^*; \theta), \quad (2)$$

Where N is the length of the output sentence, and θ is the sequence-to-sequence model parameter.

Therefore, in summary, we define our loss as below to optimize joint objective function:

$$\mathcal{L} = \mathcal{L}_M + \mathcal{L}_G. \quad (3)$$

See Figure 2 for illustration of training image-caption matching method.

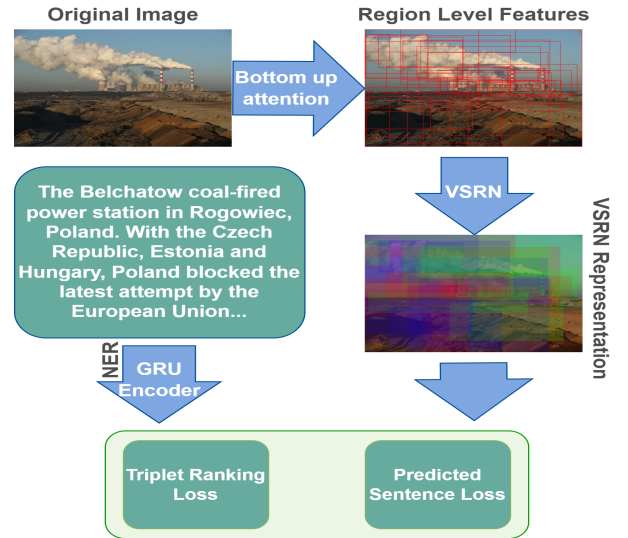


Figure 2: Training method for image-caption matching

2.1.2 Text Entailment. Based on the baseline, the SBERT-WK [15], which applies cosine similarity on two captions only, achieves 77% accuracy. After visualizing the dataset, there are many cases where SBERT-WK cannot distinguish the inverse in the semantics of two captions. Many approaches take advantage of a pre-trained NLI

task to determine the veracity of given pair of documents and claim since the output of the NLI task is the probability of corresponding class: entailment, neutral, or contradiction, which makes it practical to determine the relation between given texts. In this paper, we utilize results from pre-trained DeBERTa [9] on the MNLI dataset as additional semantic information to determine the correlation between two captions.

2.1.3 Boosting. Based on our experiences, the content of train/validation and test datasets are different. Train/validation sets only have matching or non-matching captioned images, while test sets consist of out-of-context, contradiction, or entailment captioned images. Many captioned images have one description of the image. In contrast, the other caption is the claim of news, which gave poor performance when using training on train set and evaluating directly on the test set. For this issue, we apply pre-trained various datasets to extract semantic features of the test dataset better. To combine those results, instead of finding rules for output, we use the boosting [8] technique to improve the performance. Moreover, based on the dataset's attribute, instead of boosting on training/validation data, we extract 20% of test data for boosting and evaluating the remaining 80%.

Logically, the score of the image-caption matching needs to be higher than the image-caption non-matching. Unfortunately, we recognize that there are several cases the rule we mentioned failed. To overcome these cases, we define Eq.4 towards normalizing the matching score.

$$\hat{S}(I, C) = S(I, C) - \frac{1}{2N} \sum_r [S(I, C_r) + S(I_r, C)], \quad (4)$$

where $\hat{S}$ define normalize matching score, and r define random index that satisfied $C_r \neq C$ and $I_r \neq I$. By subtracting the mean of the matching score from the N sample, the result can express the correlation degree of the given matching image-caption compared with other non-matching image-caption.

After obtaining a normalized matching score, the result of DeBERTa and SBERT-WK, we choose Transformer Encoder [5] for boosting architecture and cross-entropy loss as the objective function. See Figure 3 for illustration.

2.2 Task 2

In *Task₂*, the mission of challenge is the same to *Task₁*. However, instead of giving two captions and one image to each captioned image, the dataset of *Task₂* consists of only one caption and one corresponding image - making the task more arduous to resolve.

To settle this challenge, we adopt features of caption and the correlation between caption and image to detect the out-of-context captioned image.

For semantic features of the caption, we were fine-tuning BERT [5] on "Fake and Real News Dataset²" to detect the veracity of caption. Next, we use the trained image-text matching model in Subsection 2.1.1 to measure the correlation degree between caption and image.

Therefore, to determine if captioned image is out-of-context or not, we first compare both confidence score of not-out-of-context

²<https://www.kaggle.com/datasets/clmentbisailon/fake-and-real-news-dataset>

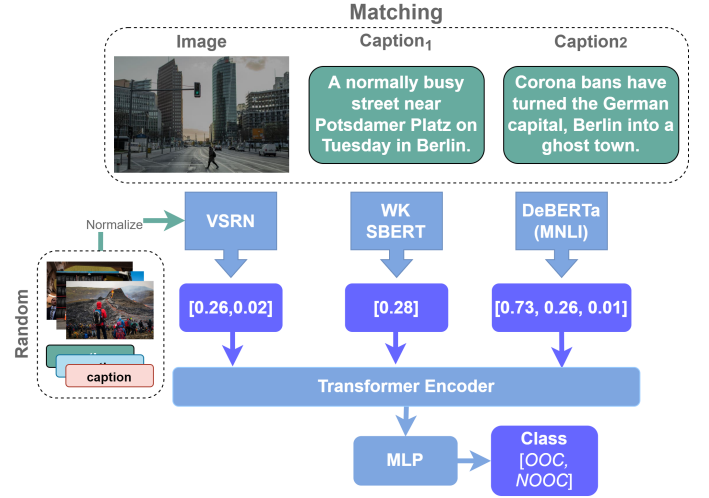


Figure 3: Illustration of boosting method. First, we extract the correlation between the image with each *caption₁* and *caption₂* base Visual Semantic Reasoning [13] and normalize it by subtracting the mean of other random images and captions. For caption semantic features, we use SBERT-WK [15] and DeBERTa (MNLI) [9] to extract cosine similarity and probability for corresponding class: (Contradiction, Neutral, Entailment). Next, we use boosting [8] with Transformer Encoder [5] to combine and improve the performance of the method.

in trained BERT model follow with two linear layer and ReLU activation for semantic features classification ($\mathbb{P}(C)$) and normalize matching score ($\hat{S}(I, C)$ (Eq.4), with threshold t_c, t_i). If $\mathbb{P}(C) < t_c$ and $\hat{S}(I, C) < t_i$, we predict captioned image as out-of-context. In the best of our experience, we choose $t_c = 0.72$ and $t_i = 0.48$. See Eq.5 and Figure 4 for more details:

$$OOC = \begin{cases} \text{True, if } \mathbb{P}(C) < t_c \text{ \& } \hat{S}(I, C) < t_i \\ \text{False, otherwise} \end{cases}, \quad (5)$$

3 EXPERIMENTAL RESULTS

To evaluate our method, we utilize the datasets and evaluation metrics of the challenge.

3.1 Implementation Details

3.1.1 Image-Text Matching Model. We set the dimension of the word embedding space to 300 and the joint embedding space of the image and caption to 1024. We start training our model on a given dataset of the grand challenge with Adam optimizer [10], learning rate 2×10^{-4} , batch size 128, margin α 0.2.

3.1.2 Boosting. For boosting method. We use two layers of Transformer Encoder [5], each containing two heads as boosting architecture. We use Adam optimizer [10], with learning rate 2×10^{-3} , weight decay 6×10^{-5} .

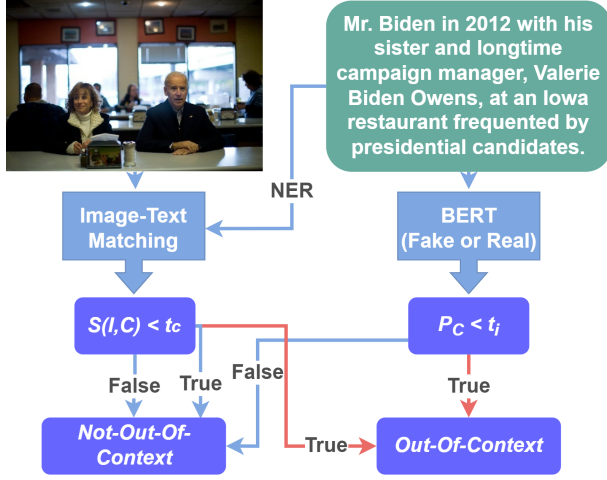


Figure 4: Illustration of the proposal for task 2. To make use of both visual and textual data, we use BERT base semantic features to classify the truthfulness of caption first. Next, we use image-text matching to measure the correlation factor between caption and image to improve the performance. If both confidence score for class not-out-of-context ($IP(C)$) and correlation measure factor ($S(I, C)$) satisfy $IP(C) < t_c$ & $\hat{S}(I, C) < t_i$, then given captioned image is out-of-context. The other else is not-out-of-context.

3.1.3 *Task 2.* We first use BERT [5] to extract semantic features. Next, we feed these features through two linear layers followed by ReLU activation to classify the out-of-context captioned image. We freeze parameters of BERT and update only parameters of two linear layers. We use binary cross-entropy loss, with Adam optimizer [10], learning rate 1×10^{-5} for training model.

3.2 Results

3.2.1 *Task 1.* For *Task*₁, we choose ANN (Linear layer with non-linear activation) and Transformer Encoder as boosting construction to compare performance. Table 1 denotes the effectiveness of our method that improves the accuracy score up to 4.25% compared to others.

Method	Accuracy(%)	Precision	Recall	F1-score
Baseline	81.25	0.8571	0.7714	0.8120
Adjust threshold[1]	83.375	0.8614	0.8142	0.8372
Boosting with ANN(ours)	83.0	0.8463	0.8261	0.8361
Boosting with Transformer Encoder(ours)	85.5	0.8502	0.8785	0.8641

Table 1: Task 1: The results

At first, we tried to boost on train/validation dataset, but the performance was poor. We have generated negative captions by appending "this is fake" or "this is not genuine" to the caption and generated additional positive captions by image captioning. Unfortunately, the performance is inferior, and the result never exceeds over 60% accuracy score.

Based on the result, we can state that we have improved the weakest aspect of the baseline and prove the effectiveness of using

global context features of images for matching caption instead of region-level features.

3.2.2 *Task 2.* Table 2 denotes the result of *Task*₂.

Method	Accuracy(%)
BERT (base semantic features)	69
BERT + Adjust threshold	72
BERT + Image-Text Matching	72
BERT + Adjust threshold + Image-Caption Matching	76

Table 2: Task 2: The results

To improve the method's performance, we first adjust the threshold for the confidence score of the BERT model. By fixing the threshold from 0.5 to 0.72, the accuracy score has improved 3%. Next, to check the effectiveness of our method, we keep the threshold of BERT base semantic features to 0.5 and adjust image-caption matching as additional information to determine the truthfulness of captioned images. The result shows that our proposal works and improves the accuracy score up to 3%. Finally, to achieve the best result, we combine these methods and achieve a 76% accuracy score. We believe this result is hard-to-beat in *Task*₂ of Grand Challenge Cheapfakes Detection.

4 CONCLUSION

In this paper, we have proposed two approaches corresponding to ACMMM'22 Grand Challenge on Detecting Cheapfakes tasks. For the first task, we focus on baseline weakness, enrich semantic information and combine them using boosting technique. For the second task, we use a simple approach that bases semantic features of caption and the correlation between image caption. We believe our results gave a competitive performance and have devoted several valuable contributions to cheapfakes detection in computer science. In the future, we will investigate creating a pre-trained model that can down-stream to different tasks and datasets. Moreover, we aim to tackle the OOC where the relationship between an image and a caption is hard to detect using only textual and visual features (e.g., paint and its title/painter) by creating a knowledge graph for different domains.

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